

PROJECT DOCUMENTATION

LabLens

Gesture-Controlled Virtual Chemistry Lab

Where Your Hand Becomes the Lab Instrument

OpenCV

MediaPipe

AI-Powered

JEE Ready

CBSE Aligned

Full Product Concept & Technical Blueprint

Inorganic Chemistry Module | Phase 1 Documentation

2025 — 2026

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Section 1: Key Idea & Vision

LabLens is a gesture-controlled virtual chemistry lab where your bare hand, tracked through a webcam, becomes the laboratory instrument. No mouse. No keyboard. No physical chemicals. Just you, the camera, and chemistry that comes alive.

Chemistry — particularly at the Class 11, 12, JEE, and BITSAT level — is one of the most visually rich subjects in science. Yet it is taught almost entirely through static textbook diagrams, chalk-and-board equations, and passive memorisation. Students who prepare for competitive examinations like JEE Advanced, JEE Mains, BITSAT, VITEEE, and SRMJEEE spend years memorising reactions, structures, trends, and exceptions — but rarely ever experience them.

LabLens bridges this gap. It creates an immersive, interactive environment where students physically perform chemistry through gesture — dipping a finger into a virtual chemical, dragging electrons into shells, navigating the periodic table with their hand, watching molecular structures assemble in 3D. Every output — every equation, every byproduct, every observation — is sourced from and cross-referenced with the exact textbooks and past papers students use for their examinations.

1.1 The Central Theme

The theme of LabLens is: **Learning by Doing, Grounded in Examination Reality**. It is not a simulation of a generic chemistry lab. It is a curated, syllabus-aligned, exam-indexed learning experience built specifically for Indian competitive examination aspirants, where every interaction is purposeful and every output is directly traceable to a source that matters to the student.

1.2 The Guiding Principles

- **Zero Passive Learning:** Every concept is accessed through an action. The student does something to learn something.
- **Syllabus Fidelity:** All content is strictly bounded by NCERT, JD Lee, and past 10 years of JEE, BITSAT, VITEEE, SRMJEEE papers. No out-of-scope content.
- **Exam Awareness:** Every piece of information carries metadata: which exam asked it, in what year, in what form, and with what frequency.
- **Visual Memory:** Molecular structures, trends, and reactions are rendered visually — because the brain retains images far better than text.
- **Progressive Depth:** The platform works equally well for a beginner exploring concepts for the first time and an advanced student doing last-minute revision.

Section 2: The Problem Being Solved

Every year, over 1.3 million students appear for JEE Mains in India. A significant proportion struggle not due to lack of effort, but due to the way chemistry is currently taught and experienced. The problems are structural:

- **Static Learning Materials:** Textbooks, even the best ones like JD Lee and NCERT, present chemistry through static text and flat diagrams. Students read about reactions but never see them unfold.
- **Rote Memorisation Culture:** Without experiential anchors, students resort to memorising hundreds of reactions and exceptions as isolated facts — leading to poor retention and application.
- **No Personalised Exam Context:** Generic chemistry apps do not map content to the specific exam a student is targeting. A BITSAT aspirant and a JEE Advanced aspirant need different emphasis.
- **Practical Labs are Inadequate:** School chemistry labs are under-resourced, limited in scope, and often not available at all in coaching centres. Students never actually perform the reactions they study.
- **3D Understanding is Missing:** Molecular geometry, orbital shapes, crystal structures — all of these are spatial concepts that students must understand from flat book diagrams. The spatial intuition never develops.
- **No Immediate Feedback Loop:** When a student reads about a reaction, there is no mechanism to immediately test understanding, see consequences, or explore variations.

LabLens directly addresses every one of these problems — by making chemistry interactive, visual, gesture-driven, and exam-indexed in a single unified platform.

Section 3: Core Interaction Design — The Gesture Lab

The defining feature of LabLens is its gesture-based interaction model. The platform uses a standard webcam — available on any laptop or external USB camera — combined with MediaPipe Hands and OpenCV to track the user's hand in real time. No special hardware is required.

3.1 How the Gesture System Works

The camera feed is processed at approximately 30 frames per second. MediaPipe Hands detects and tracks 21 landmark points on the hand — fingertips, knuckles, palm base — in real time. These landmarks are mapped to interaction zones on the screen, enabling the following gestures:

Gesture	Hand Action	Platform Response
Hover	Move hand over element/chemical on screen	Highlights element; shows preview tooltip
Dip / Tap	Bring fingertip into a chemical panel	Finger glows with the chemical's signature colour
Combine	Bring glowing finger into reaction zone or second chemical	Reaction triggered; animation plays
Pinch	Pinch index + thumb together	Zoom into periodic table block; select molecule view
Drag	Open palm move left/right	Navigate across period/group; scrub trend graph
Hold	Keep finger steady over element for 1.5s	Opens detailed character card for element
Swipe Up	Flick wrist upward	Move to next reaction / next mode
Point	Extend index finger only	Precision cursor; used in structure builder
Fist	Close hand	Exit current mode; return to home screen

3.2 The Finger Glow System

When a user dips their finger into a chemical in the side panel, an AR overlay renders a coloured glow around the fingertip — the colour corresponding to that chemical's characteristic or its visual cue in a lab context. This glow persists as long as the chemical is 'loaded' on the finger. Key behaviours:

- Each chemical has a unique glow colour — e.g. FeCl_3 glows deep amber, CuSO_4 glows blue, NaOH glows white
- If the user dips into a second chemical without reacting, the glow updates to the second chemical
- Once a reaction is triggered, a burst animation plays at the reaction zone — bubbles, colour change, precipitate formation, gas evolution — appropriate to the actual reaction
- A 'wash off' gesture (wave hand) clears the finger and resets the interaction

3.3 Reaction Zone Design

The screen is divided into three main zones: the Chemical Panel (left side — scrollable list of chemicals organised by category), the Reaction Zone (central area — a virtual beaker/flask where reactions are performed), and the Information Panel (right side — shows output, equations, observations, and exam annotations). All navigation between these zones is entirely gesture-based.

Section 4: Inorganic Chemistry Module — Full Scope

The first and primary module of LabLens covers Inorganic Chemistry comprehensively, as per the NCERT Class 11 and 12 syllabus, enriched with depth from JD Lee (Concise Inorganic Chemistry), and indexed against the past 10 years of JEE Mains, JEE Advanced, BITSAT, VITEEE, and SRMJEEE question papers.

Inorganic chemistry is not just reactions. It encompasses electronic structure, periodic trends, block properties, qualitative analysis, coordination compounds, metallurgy, industrial processes, and a rich landscape of exceptions and anomalies. LabLens provides a distinct, interactive mode for each of these domains.

4.1 Module Map

Module	Sub-areas Covered	Primary Interaction
Periodic Table Navigator	All 118 elements, blocks, periods, groups, trends	Gesture hover, pinch-zoom, hold
Electronic Configuration Builder	Aufbau, Hund, Pauli; exceptions; Cr, Cu anomalies	Drag electrons into shells
Periodic Trends Visualiser	Atomic radius, IE, EA, EN, melting point, density	Gesture-scrub 3D bar graph
s-Block Lab	Group 1 & 2 reactions, flame tests, reactivity	Finger-dip reaction lab
p-Block Lab	Groups 13-18, hybridisation, oxyacids, allotropes	Reaction + structure builder
d-Block Lab	Transition metals, colour, catalysis, variable oxidation	Reaction + colour predictor
f-Block Explorer	Lanthanides, actinides, contraction, properties	Gesture-flip flashcard + trend
Qualitative Analysis Lab	Salt analysis, group reagents, confirmatory tests	Systematic gesture-based analysis
Coordination Compounds	Ligands, CFSE, colour, magnetic, isomerism	3D structure assembler
Industrial Processes	Haber, Contact, Ostwald, Hall-Heroult	Variable control + animation
Exceptions Hall	All anomalies in inorganic compiled in one place	Gesture tap + explanation
Metallurgy	Extraction principles, ores, methods	Process flowchart navigator

Section 5: Feature Breakdown

5.1 The Living Periodic Table

The Periodic Table in LabLens is not a static reference chart. It is the central navigation hub — the home screen — from which all other modules are accessible. It is gesture-navigable, animated, and context-aware.

- Hover finger over any element → it expands to show: symbol, name, atomic number, atomic mass, electronic configuration, common oxidation states, melting/boiling point, discovery year, and a JEE frequency indicator (how many times asked in past 10 years)
- Elements are colour-coded by JEE/BITSAT exam frequency — from deep red (asked 5+ times in various forms) to pale blue (rarely tested)
- Hold gesture on element → full Character Card opens: physical properties, chemical behaviour, industrial uses, notable reactions, exceptions if any, and exam-year annotations
- Pinch gesture on a block (s, p, d, f) → table morphs to highlight only that block with its specific trends and entry point to that block's lab module
- Two-finger drag across a period → atomic radius animation plays, IE bars rise, electronegativity shading shifts — the student sees trends move as their hand moves
- Fist gesture returns to full periodic table from any sub-view
- A 'JEE lens' overlay toggle — switches element colouring from property-based to exam-frequency-based instantly

5.2 Electronic Configuration Builder

A blank atom sits in the centre of the screen. Shells appear as concentric rings. The student uses drag gestures to place electrons one by one into subshells. The system enforces all three rules — Aufbau principle (filling lower energy first), Hund's rule (maximum multiplicity), and Pauli Exclusion (no two identical quantum states). Violations glow red and trigger an explanation.

- Electrons snap to correct positions when placed correctly — satisfying visual feedback
- Once the configuration is complete, the student can hover over any subshell to see the quantum numbers of each electron
- Anomalous configurations (Cr: $[\text{Ar}] 3d^5 4s^1$, Cu: $[\text{Ar}] 3d^{10} 4s^1$) are handled with a special 'exception flag' animation — a visual burst that draws attention to the anomaly, followed by the explanation (half-filled and fully-filled stability)
- After building the configuration, the platform automatically shows the element's position on the periodic table, its block, and the JEE questions that specifically targeted this element's configuration
- A separate 'shortcut notation' view shows the noble gas core notation — which is what JEE expects in answers

5.3 Periodic Trends Visualiser

This is one of the most visually impactful features of LabLens. The periodic table rises into a 3D bar chart — each element becomes a bar whose height corresponds to the selected property's value. The student can switch properties with a gesture-select, and can drag their hand across any period or group to watch the bars highlight in sequence.

- **Atomic Radius:** Bars shrink left to right across periods; grow top to bottom in groups. Student physically sees the pattern.
- **Ionisation Energy (IE1, IE2, IE3):** The anomaly of O vs N and S vs P is visually unmistakable — those bars are shorter than expected.
- **Electron Affinity:** F vs Cl anomaly is visible — F has lower EA than Cl due to small size and electron-electron repulsion.
- **Electronegativity:** Pauling scale visualised; F at top, Cs at bottom.
- **Melting Points:** d-block shows distinctive high values; discontinuities in p-block visible.

Each anomaly in a trend is flagged with a glowing marker. When the student taps the anomaly, a pop-up explains why it occurs — and cross-references the JEE/BITSAT questions that have specifically tested that anomaly.

5.4 The Chemical Reaction Lab (Finger-Dip System)

The reaction lab is the centrepiece interaction. The Chemical Panel on the left lists all chemicals available in that block module. Chemicals are organised by type: metals, metal oxides, hydroxides, acids, salts, non-metal oxides, and gases.

- Step 1:** Student hovers finger over a chemical in the panel → name and formula appear
- Step 2:** Student dips fingertip into the chemical → fingertip begins to glow in the chemical's colour
- Step 3:** Student moves the glowing fingertip to the reaction zone (central beaker)
- Step 4:** Student dips into a second chemical OR drags a second chemical into the zone
- Step 5:** Reaction triggers — animation plays (bubbles, smoke, colour change, precipitate, flame)
- Step 6:** Right panel immediately shows: balanced chemical equation, byproducts, physical observations (colour, smell, state), conditions required (heat, catalyst, pressure), and exam annotations

The reaction database covers every reaction in the inorganic syllabus — s-block, p-block, d-block, f-block, and qualitative analysis. Each reaction entry contains:

- Reactants (with formulae and states)
- Conditions (temperature, catalyst, solvent, pressure)
- Products (with formulae, states, and colours)
- Balanced equation in molecular and ionic form
- Physical observations (colour of precipitate, colour of solution, gas evolved, smell)

- Special notes (reversibility, equilibrium position, hazard)
- Source reference (NCERT chapter, JD Lee page)
- Exam citations (which exam, which year, in what question form, marks)

5.5 Molecular & Ionic Structure Builder

After any reaction, the student can tap any product to open its 3D structure. The platform uses VSEPR theory to render the geometry. For compounds explicitly in the JEE/NCERT syllabus, hand-validated structures are used. The student can rotate the 3D structure by moving their hand.

- For p-block compounds: XeF_2 , XeF_4 , XeF_6 , ClF_3 , PCl_5 , SF_6 , H_2O , NH_3 — all rendered in 3D with bond angles labelled
- For coordination compounds: the central metal, ligands, and geometry (octahedral, tetrahedral, square planar) are shown with colour-coded bond types
- Lone pairs are visible as translucent clouds — making VSEPR intuitive rather than memorised
- Hybridisation is annotated on the central atom; students see WHY the shape is what it is
- Bond angles are displayed and the student can tap any angle to see if it deviates from ideal — and why

5.6 Qualitative Analysis (Salt Analysis) Lab

Salt Analysis is a core part of both CBSE Class 12 practical examinations and is conceptually tested in JEE. LabLens provides a full systematic qualitative analysis experience:

- A virtual 'unknown salt' appears in the beaker — the student must identify it
- Systematic analysis: preliminary tests (colour, smell, solubility), dry tests (flame, charcoal), wet tests (Group I through VI cation analysis, anion confirmatory tests)
- The student performs each test by gesture — adding reagents with finger-dip, observing results
- The system records observations and walks through the elimination logic — teaching the student how to deduce cation and anion from observations
- All group reagents are correct and match NCERT Class 12 practical manual
- At the end, the platform reveals the salt identity and shows which observations were key — reinforcing the logic

5.7 Coordination Compounds Module

Coordination compounds are among the highest-weightage topics in JEE inorganic chemistry. LabLens provides a dedicated module with the following interactions:

- Choose a central metal ion (Fe, Co, Ni, Cu, Cr, Pt etc.) with hover-select gesture
- Choose ligands from a panel — strong field (CN^- , CO, en) or weak field (H_2O , Cl^- , F^-) — and drag them toward the metal ion
- The coordination compound assembles in 3D — geometry forms based on coordination number
- Crystal Field Theory splitting diagram appears automatically — Δ_o or Δ_t is shown
- Platform predicts: colour of the complex, magnetic moment (high spin / low spin), number of unpaired electrons
- IUPAC name is generated for the assembled complex
- Isomers tab shows all possible isomers (geometrical, optical, linkage, ionisation) with 3D structures
- JEE past questions on this exact compound or compound type are listed in the annotations panel

5.8 Exceptions Hall

The Exceptions Hall is a dedicated section that compiles every anomaly, exception, and counterintuitive result in inorganic chemistry. These are the facts that repeatedly appear in JEE and BITSAT precisely because they test conceptual depth rather than surface memorisation.

Exception	What It Says	Why It Happens	Exam Frequency
Cr and Cu config	Cr is $[\text{Ar}]3d^5 4s^1$, Cu is $[\text{Ar}]3d^{10} 4s^1$	Half-filled/fully-filled d subshell stability	Very High - JEE, BITSAT
IE of O < N	Despite higher nuclear charge, O has lower IE than N	Half-filled $2p^3$ first IE is extra stable	Very High - JEE Advanced
EA of F < Cl	F has lower electron affinity than Cl	Small size causes e^- - e^- repulsion	High - JEE, BITSAT
Diagonal Relationship	Li-Mg, Be-Al, B-Si show similar properties	Charge density similarity across diagonal	Medium - JEE Mains
Inert Pair Effect	Heavier p-block elements prefer lower oxidation states	Reluctance to use s-orbitals in bonding	High - JEE, BITSAT
Anomalous H_2O properties	High boiling point, maximum density at 4°C	Extensive H-bonding; density anomaly	Medium - JEE, BITSAT
NO is paramagnetic	NO has an odd electron, making it paramagnetic	Unpaired electron in π^* orbital	Medium - JEE Advanced

Each exception in the hall is accessed with a tap gesture. The student sees: the exception stated clearly, the deep explanation, a visual (orbital diagram, energy diagram, or graph), and the specific JEE/BITSAT questions where this exception was the key to solving the problem.

Section 6: Learning Curve & Pedagogical Design

LabLens is designed with four progressive learning layers. A student can enter at any layer depending on their current level, and can move between layers within a single session. The design philosophy is: curiosity first, structure second, exam readiness third, mastery fourth.

Layer 1 — Explore Mode (Curiosity First)

Free, unguided interaction. The student opens the platform and can do anything — navigate the periodic table, dip into any chemical, build any configuration. There are no objectives, no scores, no time limits. This layer is designed to trigger intrinsic curiosity and make the student comfortable with the gesture interface. It functions like a sandbox chemistry game.

- All chemicals and elements accessible immediately
- Every reaction that occurs in this mode is logged in the student's personal 'Reaction History'
- Interesting or unexpected reactions produce a 'Discover' badge — collecting these rewards exploration

Layer 2 — Guided Mission Mode (Structured Learning)

Structured like game levels. The platform generates missions aligned with the NCERT chapter sequence and exam importance. Examples:

- 'Complete all reactions of Group 1 metals with water and acids'
- 'Build the electronic configurations of all d-block elements in Period 4'
- 'Identify the cation and anion in 5 unknown salts using qualitative analysis'

Missions have progress bars, unlock subsequent missions, and produce a chapter completion certificate. The sequencing follows the NCERT chapter order but prioritises by JEE weightage within each chapter.

Layer 3 — Exam Lens Mode (Exam-Ready Filtering)

The same lab, same chemicals, same interactions — but now every output is filtered and presented through the lens of the student's target exam. The student selects their exam mode at the top:

- **JEE Advanced:** Deep conceptual explanations, multi-step reaction mechanisms, coordination compound depth, exception analysis
- **JEE Mains:** Balanced coverage of all chapters, factual reactions, standard structures, naming conventions
- **BITSAT:** Speed-oriented; highlights frequently repeated questions; short-answer format annotations
- **VITEEE / SRMJEEE:** Syllabus is slightly different; platform adjusts which content is highlighted and which is marked 'out of scope'

- **CBSE Boards:** NCERT-first; laboratory observations; salt analysis; balanced equations as per board convention

Layer 4 — AI Examiner Mode (Socratic Dialogue)

After the student performs any interaction — a reaction, a configuration, a trend navigation — the AI Examiner activates. It does not quiz. It conducts a Socratic dialogue. The AI is strictly bounded by the platform's knowledge database — it cannot answer anything outside the syllabus, preventing hallucination.

Example: Student just reacted FeCl_3 with NaOH . AI Examiner says: "You just observed a reddish-brown precipitate. JEE 2019 Paper 1 asked: What is the colour and formula of the product? What is the systematic IUPAC name of this compound? Think for a moment before the answer reveals." Then it waits, hints if needed, reveals, and explains the underlying chemistry.

Section 7: Knowledge Sources & Database Architecture

7.1 Knowledge Sources

Source	Type	Role in Platform	Scope
NCERT Class 11 Chemistry	Primary Textbook	Foundational concept definitions, standard reactions	Chapters 1-4, Chapter structure
NCERT Class 12 Chemistry	Primary Textbook	Coordination, d-block, f-block, metallurgy, Qualitative Analysis	Chapters 5-9
JD Lee — Concise Inorganic Chemistry	Reference Book	Deeper explanations, exceptions, non-trivial properties	Full book, literature and JEE examples
JEE Mains Past Papers (2014-2024)	Question Database	Reaction citations, question form, marks, frequency of topics	10 years of papers
JEE Advanced Past Papers (2014-2024)	Question Database	Advanced concepts, multi-concept questions, problem papers	10 years of papers
BITSAT Past Papers (2014-2024)	Question Database	Speed-format questions, topic distribution	10 years, BITSAT-specific emphasis
VITEEE Past Papers (2014-2024)	Question Database	VIT syllabus scope, question style, subject distribution	10 years
SRMJEEE Past Papers (2014-2024)	Question Database	SRM question style and topic preferences	10 years

7.2 Database Schema — Reactions Table

Every reaction is stored as a structured record with the following fields:

Field	Type	Description
reaction_id	UUID	Unique identifier for the reaction
reactant_1	String	Formula and name of first reactant
reactant_2	String	Formula and name of second reactant (if any)
conditions	JSON	Temperature, catalyst, solvent, pressure required
products	JSON Array	All products with formula, state, colour, smell
balanced_equation	String	Full balanced molecular equation (HTML formatted)
ionic_equation	String	Net ionic equation where applicable
observations	JSON	Colour change, precipitate, gas, heat evolved
block_category	Enum	s_block / p_block / d_block / f_block / general
ncert_chapter	String	NCERT chapter reference
jd_lee_page	Integer	JD Lee page number for deeper reading

Field	Type	Description
exam_citations	JSON Array	List of {exam, year, question_type, marks}
jee_frequency	Integer	Number of times this reaction appeared in JEE (10yr)
bitsat_frequency	Integer	Number of times appeared in BITSAT (10yr)
difficulty_level	Enum	easy / medium / hard / advanced
glow_color_hex	String	Hex colour for finger glow of primary reactant
animation_type	Enum	precipitate / gas / colour_change / exothermic / none

7.3 Database Schema — Elements Table

Field	Type	Description
atomic_number	Integer	Primary key
symbol	String	Element symbol (e.g. Fe, Cu, Na)
name	String	Full IUPAC name
atomic_mass	Float	Standard atomic weight
electronic_config	String	Full configuration (HTML superscript formatted)
block	Enum	s / p / d / f
period	Integer	Period number (1-7)
group	Integer	Group number (1-18)
oxidation_states	JSON Array	All known oxidation states with most common flagged
atomic_radius_pm	Float	Covalent atomic radius in picometres
ionic_radius	JSON	Ionic radii for each common oxidation state
ie1_kj_mol	Float	First ionisation energy
ie2_kj_mol	Float	Second ionisation energy
electron_affinity	Float	First electron affinity
electronegativity	Float	Pauling electronegativity
melting_point_K	Float	Melting point in Kelvin
boiling_point_K	Float	Boiling point in Kelvin
exceptions	JSON Array	List of anomalies/exceptions for this element
jee_frequency	Integer	Total appearances in JEE (10yr, all forms)
character_card	JSON	Full display card data: uses, fun fact, key reactions

Field	Type	Description
glow_color_hex	String	Colour for periodic table highlight

Section 8: Exam Relevance & Question Mapping

A core differentiator of LabLens is that every piece of content is tagged to examination data. This means the student always knows not just what a concept is, but why it matters for their exam — and exactly how it has been asked before.

8.1 Topic-wise JEE Weightage (Inorganic — Past 10 Years)

Topic	JEE Mains Avg Q/Yr	JEE Adv Avg Q/Yr	BITSAT Avg Q/Yr	Priority
d and f Block Elements	2-3	1-2	2-3	VERY HIGH
Coordination Compounds	2-3	2	2-3	VERY HIGH
p-Block Elements (Gp 15-18)	2-3	1-2	2-3	VERY HIGH
p-Block Elements (Gp 13-14)	1-2	1	1-2	HIGH
General Principles of Extraction	1	1	1	HIGH
Hydrogen and its Compounds	1	0-1	1	MODERATE
s-Block Elements	1	0-1	1	MODERATE
Periodic Table & Periodicity	1-2	1	2	HIGH
Chemical Bonding (Inorganic)	2-3	2	2-3	VERY HIGH
Qualitative Analysis	0-1	0-1	1	MODERATE

8.2 Question Type Distribution by Exam

Each exam tests inorganic chemistry in a different format. LabLens's Exam Lens mode adapts the information panel to show content in the exact format the student will encounter in their exam:

Question Type	JEE Mains	JEE Advanced	BITSAT	CBSE Boards
Product identification from reaction	Yes — MCQ	Yes — MCQ/Integer	Yes — MCQ	Yes — 2-3 marks
Balance the equation	Rarely	Rarely	Rarely	Yes — 2 marks
Observation-based (colour, ppt)	Yes — MCQ	Yes	Yes	Yes — practical
Exception / anomaly questions	Yes	Yes — multi-concept	Yes	Rarely
Structure / geometry prediction	Yes — MCQ	Yes — matching	Yes	Yes
IUPAC naming	Yes	Yes	Yes	Yes
Electronic config anomalies	Yes	Yes	Yes	Yes — 1 mark

Question Type	JEE Mains	JEE Advanced	BITSAT	CBSE Boards
Industrial process conditions	Yes	Yes	Yes	Yes — 3 marks
Matching / multiple correct	No (Mains)	Yes	No	No
Integer / numerical answer	Yes (Mains)	Yes	No	No

8.3 Sample Exam Annotation (How It Appears In Platform)

When a student reacts KMnO_4 with FeSO_4 in acidic medium, the right panel shows:

- Balanced equation: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$
- Observation: Purple colour of KMnO_4 decolourises; solution turns light pink then colourless
- Change in oxidation state: Mn goes from +7 to +2 (reduction); Fe goes from +2 to +3 (oxidation)
- JEE Mains 2022: Asked which ion acts as reducing agent in this reaction [Answer: Fe^{2+}]
- JEE Mains 2019: Asked the n-factor of KMnO_4 in acidic medium [Answer: 5]
- BITSAT 2021: Asked the colour change observed when KMnO_4 is reduced in acidic medium
- JEE Advanced 2017: Used this reaction as part of a back-titration problem

Section 9: Technology Stack

LabLens is built entirely on open-source and freely available tools, making it accessible to develop and deploy without commercial licensing costs. The stack is Python-first, with a lightweight frontend layer.

Layer	Technology	Purpose
Hand Tracking	MediaPipe Hands (Google)	Real-time 21-point hand landmark detection at 30fps
Computer Vision	OpenCV (Python)	Camera input, frame processing, AR overlay rendering
AR Finger Glow	OpenCV + NumPy	Custom colour overlay on fingertip coordinates from MediaPipe
Reaction Animation	Pygame / OpenCV draw	Bubble, precipitate, colour-change, smoke animations
3D Molecular Structures	Py3Dmol + RDKit	Generate and render 3D structures from SMILES notation
2D Structural Formulae	RDKit	2D Lewis structure rendering for display panel
Reaction Database	PostgreSQL / SQLite	Stores all reactions, elements, compounds with full metadata
Exam Citation Engine	Python backend	Queries and returns exam-specific annotations for each reaction
AI Examiner	RAG + LLM (local or API)	Retrieval-augmented generation bounded to platform knowledge base
Knowledge Base (RAG)	FAISS + LangChain	Vector store of all textbook content + past paper questions
Frontend UI	Pygame or Electron	Screen layout, panels, gesture zone detection, animation rendering
Session Tracking	Python + SQLite	Records student interactions, progress, mission completion
Configuration Data	JSON files	Electronic config data, exception explanations, character cards

Critical Design Choice: The AI Examiner is grounded via Retrieval-Augmented Generation (RAG). It can only retrieve information from the platform's own curated database — it cannot generate answers from general web knowledge. This ensures 100% syllabus fidelity and eliminates hallucination risk for exam-context responses.

Section 10: System Architecture

The system is structured in four distinct layers that work in concert to deliver the full platform experience:

Layer A — Input & Perception

Webcam feed → OpenCV frame capture → MediaPipe Hand landmark detection → Gesture classifier (rule-based + ML) → Gesture event dispatcher

Layer B — Interaction Logic

Gesture events → Interaction Manager → determines current mode (Periodic Table / Reaction Lab / Config Builder / Trend Visualiser) → routes to appropriate module handler → state update

Layer C — Data & Knowledge

Module handler → queries Reaction DB / Element DB / Exam Citation Engine → if AI Examiner active: query → FAISS vector store → RAG retrieval → LLM generation → response

Layer D — Output & Rendering

Data response → Animation Engine (reaction animations) → Structure Renderer (3D/2D molecules) → Information Panel (equations, observations, exam tags) → AR Overlay (finger glow) → composite frame → display

10.1 Data Flow Diagram (Reaction Lab)

Step	Component	Input	Output
1	Webcam	Physical world	Raw video frame
2	OpenCV	Raw frame	Processed frame (BGR)
3	MediaPipe Hands	Processed frame	21 hand landmarks (x, y, z)
4	Gesture Classifier	Landmarks	Gesture event (e.g. 'finger_dip:FeCl3')
5	Interaction Manager	Gesture event	Action: 'load_chemical FeCl3 on finger'
6	AR Overlay Engine	Action + finger coords	Glow rendered on fingertip
7	Reaction Trigger	Two chemicals loaded	Query: reaction(FeCl3, NaOH)
8	Reaction DB	Query	Reaction record JSON

Step	Component	Input	Output
9	Animation Engine	Reaction type (precipitate)	Reddish-brown ppt animation
10	Info Panel Renderer	Reaction record + Exam mode	Equation, observations, JEE tags displayed
11	AI Examiner (opt)	Reaction record	Socratic prompt generated via RAG
12	Display Compositor	All rendered layers	Final frame shown to student

Section 11: Future Extensions

LabLens is architected as a modular platform. The gesture interaction system, the AI Examiner, the examination engine, and the rendering pipeline are all subject-agnostic. Once the Inorganic Chemistry module is complete and validated, the same framework can be extended to:

11.1 Organic Chemistry Module

Sources: MS Chauhan, NCERT Class 12 Part 1, past 10 years JEE/BITSAT

- Named Reaction Theatre: Student performs gesture to add reagents to a substrate; named reaction (Aldol, Cannizzaro, Sandmeyer, Reimer-Tiemann etc.) triggers with mechanism animation
- Functional Group Identification Lab: Student tests an unknown compound with various reagents (Lucas test, Tollens, Fehling, DNP) — identifies the functional group from observations
- Mechanism Visualiser: Arrow-pushing mechanisms shown as animated electron flow; student can step through each step with gesture
- IUPAC Naming Trainer: Build a carbon skeleton with gesture; platform generates the IUPAC name and shows rules applied
- Isomer Explorer: Generate all structural, geometrical, and optical isomers of a given formula — visualise in 3D
- Reaction Map: A network graph of all organic transformations — navigate from one compound class to another, seeing all possible transformations

11.2 Physical Chemistry Module

Sources: OP Tandon, NCERT, past 10 years JEE/BITSAT

- Electrochemistry Cell Builder: Gesture-assemble galvanic / electrolytic cells; watch current flow, calculate EMF with Nernst equation rendered in real time
- Chemical Kinetics Graph: Adjust concentration, temperature, catalyst with gestures; watch rate vs time curves update live
- Equilibrium Shifter: Le Chatelier's Principle made interactive — change conditions, watch equilibrium position shift with animation
- Thermodynamics Engine: Gesture-control a thermodynamic process (isothermal, adiabatic, isobaric); watch P-V curves, calculate work done, heat, internal energy change
- Colligative Properties Lab: Dissolve solute in solvent with gesture; observe elevation of boiling point, depression of freezing point, osmotic pressure in real time

11.3 Physics Module

- Ray Optics Lab: Draw mirrors and lenses with gesture; shoot light rays; watch reflection, refraction, image formation
- Circuit Builder: Gesture-place resistors, capacitors, inductors; connect with gesture-drawn wires; measure voltage and current
- Wave & Oscillation Visualiser: Adjust amplitude, frequency, phase with gestures; watch superposition, interference, standing waves
- Mechanics Simulator: Projectile motion, circular motion, SHM — set initial conditions with gesture, watch trajectory

11.4 Biology Module

- Cell Interaction Lab: Navigate into a cell; tap organelles to see their function; watch processes like protein synthesis, ATP generation
- Genetics Punnett Square Builder: Gesture-drag alleles into a cross; watch inheritance pattern emerge
- Enzyme Kinetics: Visualise substrate-enzyme interaction, Michaelis-Menten curve, inhibitor effects

11.5 Platform-Level Expansions

- Mobile Version: Port the gesture system to smartphone front camera using MediaPipe on Android/iOS
- Multiplayer Lab: Two students collaborate in the same virtual lab via network — one loads a chemical, the other adds the second; compete in reaction speed challenges
- Teacher Dashboard: Teachers set missions, monitor student progress, see which reactions/exceptions students are struggling with, generate class-level analytics
- AR Glasses Version: Port to Meta Quest or similar device for a fully immersive lab environment
- Accessibility Mode: Voice-gesture hybrid for students with limited hand mobility

Section 12: Scope & Impact

12.1 Who Is This For?

User Group	How LabLens Serves Them
JEE Mains Aspirants (Class 12)	Full syllabus coverage, question-format annotation, reactions indexed to JEE frequency, exceptions highlighted
JEE Advanced Aspirants	Deep conceptual exploration, multi-step reasoning prompts, coordination compound depth, anomaly analysis
BITSAT Aspirants	Speed-optimised mode, repeated question patterns highlighted, concise answer format
VITEEE / SRMJEEE Aspirants	Syllabus-filtered content, exam-specific topic emphasis flags
CBSE Class 12 Board Students	NCERT-first content, practical-oriented lab (salt analysis), balanced equation format per board guidelines
Teachers & Coaching Faculties	Demonstration tool for classroom; visualise trends, reactions, and structures on projector via gesture
Self-Study Students	Explore mode for independent learning without needing a teacher or institute

12.2 What Makes LabLens Genuinely New?

- **Gesture as Primary Interface:** No existing chemistry EdTech platform uses hand gesture as the primary and only interaction mode. Every competitor uses mouse, keyboard, or touch.
- **Finger-Chemical Colour Glow:** The finger glow metaphor — where your fingertip literally carries the chemical — is a novel interaction design that creates physical memory of the reaction.
- **Syllabus-Bounded AI:** The AI Examiner can only answer from the platform's curated knowledge base. This is fundamentally safer and more reliable than a general-purpose chatbot for exam preparation.
- **Exam Citation on Every Output:** No other platform tells you — in real time, as you perform a reaction — exactly which exam asked this, in which year, in what form. This is a unique value proposition.
- **Inorganic Beyond Reactions:** Competitors that simulate reactions ignore the other 60% of inorganic: trends, configurations, structures, exceptions, qualitative analysis. LabLens covers all of it.
- **Zero Hardware Cost:** The only hardware required is a standard laptop webcam. No proprietary controllers, no VR headsets, no sensors. This makes LabLens accessible to any student anywhere.

12.3 Measurable Impact Targets

Metric	Target
Reactions covered in Phase 1 (Inorganic)	400+ unique reactions from NCERT + JD Lee
Elements with full character cards	All 118 elements

Metric	Target
Past exam questions indexed	1,000+ questions from JEE + BITSAT + VITEEE + SRMJEEE (10 years)
Exceptions and anomalies catalogued	50+ inorganic exceptions with explanations and exam citations
Qualitative analysis scenarios	30+ unknown salt scenarios with varying cation/anion combinations
3D structures available	100+ compounds with VSEPR-based 3D rendering
Learning modes	4 progressive layers (Explore, Mission, Exam Lens, AI Examiner)
Exam formats supported	JEE Mains, JEE Advanced, BITSAT, VITEEE, SRMJEEE, CBSE Board
Hardware requirement	Any laptop or desktop with a standard webcam; no additional hardware

LabLens represents a fundamentally new way to learn and interact with chemistry — where the depth of a JD Lee is made accessible through the intuitiveness of gesture, and where every chemical interaction carries the weight of real examination history. It is built not to replace the textbook, but to make the textbook come alive.

End of Document — LabLens Project Blueprint v1.0
Inorganic Chemistry Module | Phase 1